Strategic Operations for Gourmet Wafers

Forecasting and Demand Planning

To ensure Gourmet Wafers scales effectively, we apply a 20% annual growth factor to a seasonally indexed baseline. Using last year's average of 1,625 boxes per month, we forecast a next-year monthly average of 1,950 boxes. However, because our December seasonality index is 1.97, we must plan for a peak demand of 3,842 boxes. This data-driven approach allows us to translate market trends into a daily production requirement of approximately 2,096 cookies during the holiday rush, ensuring we capture maximum revenue without missing customer orders.

Capacity Planning and Process Selection

Our capacity strategy balances artisan quality with operational efficiency. With an oven capacity of 3,840 cookies per 8-hour shift, a single oven is mathematically sufficient for our peak daily demand of 2,096 cookies. However, we will implement a semi-automated process for portioning and topping. This decision ensures each cookie maintains a weight of $25\text{g} \pm 2\text{g}$, reducing ingredient "giveaway" and maintaining the brand's gourmet standards while providing a 20% capacity buffer for surges or equipment downtime.

Inventory and Supply Management

The reliability of our supply chain is anchored by our Reorder Point (ROP) for critical ingredients like pecans. By calculating a safety stock based on a 95% service level ($z \approx 1.65$), we set our ROP at 131 kg. This means when pecan stock hits this level, we automatically trigger a new order from our primary, high-reliability supplier. This ensures that even with a 7-day lead time and demand variability, the production line never stops, which is essential during our high-volume December period.

Facility Layout and Job Design

To minimize waste and maintain hygiene, we will utilize a product-oriented line flow layout. This arrangement moves the product linearly from mixing to baking, cooling, and finally packaging. Our work measurement study indicates that one batch of 240 cookies requires 55 minutes of direct labor. During peak periods, this requires a staffing level of two full-time equivalents (FTEs) to handle the 10 batches needed per day, allowing workers to specialize in specific tasks like quality control and packaging to maximize throughput.

Quality Management and Product Design

Quality management for a family recipe requires rigid standards to protect brand equity. We will implement an attribute sampling plan, inspecting 10 cookies per batch against a "golden brown" reference chart and a 100% pecan-placement requirement. If a sample exceeds a 2% defect rate, the entire batch is flagged for rework. This strict quality control is paired with a product

design strategy that locks in the flagship recipe while allowing for high-margin variants, such as premium holiday tins, to increase the average transaction value during peak seasons.

Location Analysis and Scheduling

Our location strategy focuses on proximity to our primary demand clusters in the Massachusetts area to minimize shipping costs and lead times. By locating near our customer base, we can save an estimated $2 per box in logistics costs. This proximity supports our scheduling decision to run a single, tight 8-hour shift in December, where we bake 10 batches daily. This schedule provides enough "slack" time for equipment cleaning and administrative tasks, ensuring the facility operates at peak efficiency without incurring excessive overtime costs.